

Introduction

In this lab, we explore gRPC (Google Remote Procedure Call) to establish seamless communication between multiple rover clients and a centralized ground control server. Using gRPC's high-performance, language-agnostic framework, we simulate a fleet of 10 autonomous rovers navigating a 2D terrain, executing movement commands, and interacting with mines.

The lab is divided into three main parts:

1. Defining Protocol Buffers – Creating the .proto file that specifies the gRPC services and message formats.
2. Implementing the Server – Developing a Ground Control server that manages maps, commands, and mine data while handling rover requests.
3. Implementing the Clients – Programming rover clients to communicate with the server, execute commands, disarm mines, and report their mission status.

This lab builds on concepts from Lab 1, extending them with real-time client-server interaction using gRPC streaming. The primary goal is to understand how remote procedure calls enable efficient communication in distributed systems, a crucial skill for modern software engineering.

Results

Result of when the rover just explodes after stepping on a mine and tries to move away.

Client

```
PS C:\Users\charl\OneDrive\Documents\COEtrial\892lab2> python client.py 1
Rover 1 received map of size 4x3
Rover 1 executing command: M
Rover 1 moved to position: (1, 0)
Rover 1 executing command: M
Rover 1 moved to position: (2, 0)
Rover 1 has landed on an active mine at (2, 0)!
Rover 1 executing command: M
Rover 1 is leaving an active mine without disarming at (2, 0)!
Server status acknowledgement: Status received
Rover 1 path written to path_1.txt
```

Server

```
PS C:\Users\charl\OneDrive\Documents\COEtrial\892lab2> python server.py
Ground Control gRPC server started on port 50051.
Rover 1 reported status: FAILURE
█
```

Result of when the rover disarmed a mine, never blew up, and just ran out of moves.

Client

```
PS C:\Users\charl\OneDrive\Documents\COETrial\892lab2> python client.py 2
Rover 2 received map of size 4x3
Rover 2 executing command: M
Rover 2 moved to position: (1, 0)
Rover 2 executing command: R
Rover 2 executing command: D
Rover 2 attempting to disarm mine at position (1, 0)
No active mine to disarm at current position.
Rover 2 executing command: R
Rover 2 executing command: M
Rover 2 moved to position: (0, 0)
Rover 2 executing command: M
Movement out of bounds! Staying at the same position.
Rover 2 executing command: L
Rover 2 executing command: M
Movement out of bounds! Staying at the same position.
Rover 2 executing command: L
Rover 2 executing command: R
Rover 2 executing command: D
Rover 2 attempting to disarm mine at position (0, 0)
No active mine to disarm at current position.
Rover 2 executing command: R
Rover 2 executing command: L
Rover 2 executing command: L
Rover 2 executing command: M
Rover 2 moved to position: (1, 0)
Rover 2 executing command: R
Rover 2 executing command: L
Rover 2 executing command: L
Rover 2 executing command: M
Rover 2 moved to position: (1, 1)
Rover 2 executing command: M
Rover 2 moved to position: (1, 2)
Rover 2 executing command: R
```

```
Rover 2 executing command: D
Rover 2 attempting to disarm mine at position (1, 2)
No active mine to disarm at current position.
Rover 2 executing command: L
Rover 2 executing command: M
Movement out of bounds! Staying at the same position.
Rover 2 executing command: L
Rover 2 executing command: D
Rover 2 attempting to disarm mine at position (1, 2)
No active mine to disarm at current position.
Rover 2 executing command: M
Rover 2 moved to position: (0, 2)
Rover 2 executing command: L
Rover 2 executing command: D
Rover 2 attempting to disarm mine at position (0, 2)
No active mine to disarm at current position.
Rover 2 executing command: L
Rover 2 executing command: D
Rover 2 attempting to disarm mine at position (0, 2)
No active mine to disarm at current position.
Rover 2 executing command: L
Rover 2 executing command: R
Rover 2 executing command: L
Rover 2 executing command: M
Movement out of bounds! Staying at the same position.
Rover 2 executing command: R
Rover 2 executing command: L
Rover 2 executing command: M
Movement out of bounds! Staying at the same position.
```

```
Rover 2 executing command: M
Movement out of bounds! Staying at the same position.
Rover 2 executing command: R
Rover 2 executing command: M
Movement out of bounds! Staying at the same position.
Rover 2 executing command: R
Rover 2 executing command: L
Rover 2 executing command: M
Movement out of bounds! Staying at the same position.
Rover 2 executing command: L
Rover 2 executing command: M
Movement out of bounds! Staying at the same position.
Rover 2 executing command: L
Rover 2 executing command: M
Rover 2 moved to position: (0, 1)
Rover 2 has landed on an active mine at (0, 1)!
Rover 2 executing command: D
Rover 2 attempting to disarm mine at position (0, 1)
Rover 2 received mine serial: tapsgyjqd1
Rover 2 computed mine PIN: 1692127
Server acknowledgement: PIN received
Rover 2 executing command: M
Rover 2 moved to position: (0, 0)
Rover 2 executing command: R
Rover 2 executing command: L
Rover 2 executing command: R
Rover 2 executing command: M
Movement out of bounds! Staying at the same position.
Rover 2 executing command: M
Movement out of bounds! Staying at the same position.
Rover 2 executing command: M
Movement out of bounds! Staying at the same position.
Rover 2 executing command: M
```

```
Movement out of bounds! Staying at the same position.
Rover 2 executing command: M
Movement out of bounds! Staying at the same position.
Rover 2 executing command: M
Movement out of bounds! Staying at the same position.
Rover 2 executing command: M
Movement out of bounds! Staying at the same position.
Rover 2 path written to path_2.txt
```

Server

```
PS C:\Users\charl\OneDrive\Documents\COEtrial\892lab2> python server.py
Ground Control gRPC server started on port 50051.
Rover 1 reported status: FAILURE
Rover 2 shared mine PIN: 1692127
Rover 2 reported status: SUCCESS
█
```

Result of when the rover disarmed a mine, but ended up hitting another mine.

Client

```
PS C:\Users\charl\OneDrive\Documents\COETrial\892lab2> python client.py 10
Rover 10 received map of size 4x3
Rover 10 executing command: M
Rover 10 moved to position: (1, 0)
Rover 10 executing command: M
Rover 10 moved to position: (2, 0)
Rover 10 has landed on an active mine at (2, 0)!
Rover 10 executing command: D
Rover 10 attempting to disarm mine at position (2, 0)
Rover 10 received mine serial: bhh7zc2pp1
Rover 10 computed mine PIN: 11113240
Server acknowledgement: PIN received
Rover 10 executing command: M
Rover 10 moved to position: (3, 0)
Rover 10 executing command: M
Movement out of bounds! Staying at the same position.
Rover 10 executing command: L
Rover 10 executing command: M
Rover 10 moved to position: (3, 1)
Rover 10 executing command: L
Rover 10 executing command: R
Rover 10 executing command: L
Rover 10 executing command: L
Rover 10 executing command: R
Rover 10 executing command: M
Rover 10 moved to position: (2, 1)
Rover 10 executing command: R
Rover 10 executing command: M
Rover 10 moved to position: (2, 2)
Rover 10 executing command: M
Movement out of bounds! Staying at the same position.
Rover 10 executing command: L
Rover 10 executing command: M
```

```
Rover 10 moved to position: (1, 2)
Rover 10 executing command: R
Rover 10 executing command: D
Rover 10 attempting to disarm mine at position (1, 2)
No active mine to disarm at current position.
Rover 10 executing command: M
Movement out of bounds! Staying at the same position.
Rover 10 executing command: L
Rover 10 executing command: M
Rover 10 moved to position: (0, 2)
Rover 10 executing command: L
Rover 10 executing command: M
Rover 10 moved to position: (0, 1)
Rover 10 has landed on an active mine at (0, 1)!
Rover 10 executing command: L
Rover 10 executing command: M
Rover 10 is leaving an active mine without disarming at (0, 1)!
Rover 10 path written to path_10.txt
```

Server

```
PS C:\Users\charl\OneDrive\Documents\COETrial\892lab2> python server.py
Ground Control gRPC server started on port 50051.
Rover 1 reported status: FAILURE
Rover 2 shared mine PIN: 1692127
Rover 2 reported status: SUCCESS
Rover 10 shared mine PIN: 11113240
Rover 10 reported status: FAILURE
█
```

Conclusion

By completing this lab, we successfully implemented a real-time, distributed communication system using gRPC. The Ground Control server acted as a centralized hub, handling multiple rover clients that operated independently, retrieving maps, executing movement commands, and disarming mines.

Through this lab, we gained hands-on experience with:

- Defining and using Protocol Buffers (.proto files) for structured communication.
- Building a gRPC server to handle multiple concurrent clients.
- Streaming data efficiently between server and clients.
- Managing remote procedure calls for request-response interactions.

This exercise demonstrates the power of gRPC in distributed applications, providing a solid foundation for building scalable, high-performance communication systems. The concepts learned here are directly applicable to fields like autonomous robotics, IoT, cloud computing, and microservices architecture.